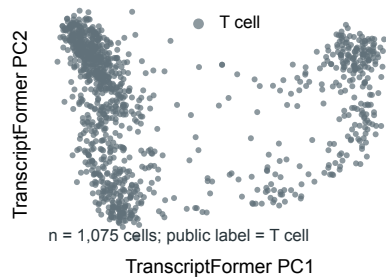
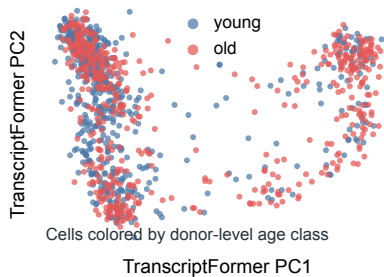


Cellular landscape, cohort structure and sources of confounding

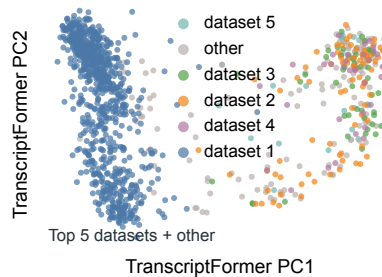
A Embedding projection: cell annotation



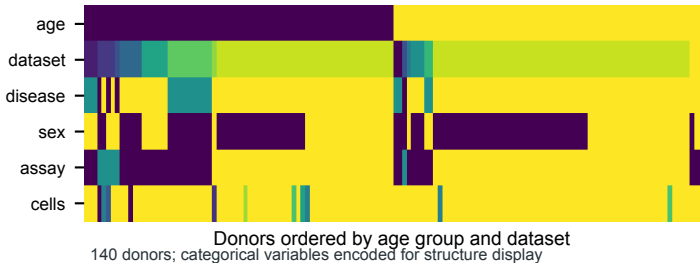
B Donor-level age class



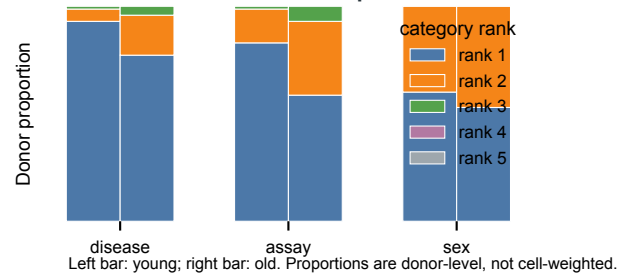
C Study-origin structure



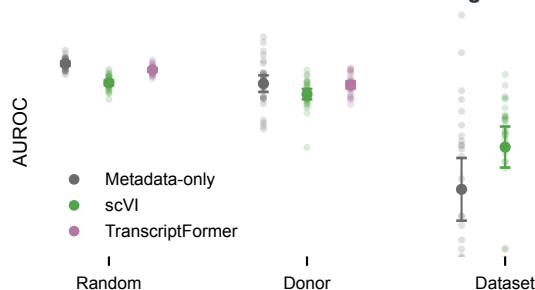
D Donor-level metadata structure



E Donor-level metadata composition



F Performance across validation strategies



G Embedding AUROC minus metadata-only

